

# STX882 & SRX882 Hardware Guide

## 433 MHz RF Triggers & Pico Breadboard Integration

### 1. Overview & Benefits

Upgrading from entry-level SMAKN (FS1000A / MX-05V) modules to the STX882 (transmitter) and SRX882 (receiver) is the single most effective hardware upgrade you can make to improve the range and reliability of your 433 MHz RF triggers. Because both hardware sets use ASK/OOK modulation, your software scripts (receive.py, send.py, pico\_bridge.py) work out of the box.

- Superheterodyne Receiver (SRX882): Uses a crystal oscillator to filter out background static, providing a clean data line, lowering CPU usage, and offering -107 dBm sensitivity.
- High Transmit Power (STX882): Transmits up to 100mW at 5V, allowing signals to penetrate walls and reach farther rooms.

### 2. Raspberry Pi Pico Breadboard Guide

A breadboard is used to prototype circuits without soldering. Here is how to mount the Pico:

1. Straddle the Center Divider:  
Place the Raspberry Pi Pico lengthwise down the middle of the breadboard so the center divider splits the left row of pins and the right row of pins. Each Pico pin will now have 4 separate connection holes in its horizontal row.
2. Establish the 3.3V Power Rail:  
Connect a jumper wire from Pin 36 (3V3 OUT) of the Pico to the red '+' power bus strip on the side of the breadboard. This turns that entire rail into a 3.3V power source.
3. Establish the Common Ground Rail:  
Connect a jumper wire from Pin 38 (GND) of the Pico to the blue '-' ground bus strip on the side of the breadboard. This turns the entire rail into a ground return path.

### 3. Pico RF Wiring Tables

**WARNING: The Raspberry Pi Pico's GPIO pins are NOT 5V tolerant (3.3V max). You must run the SRX882 receiver at 3.3V to protect the Pico, but run the STX882 transmitter at 5V for maximum range.**

#### STX882 Transmitter (Square Board)

| STX882 Pin | Pico Pin # | Signal Name | Purpose                   |
|------------|------------|-------------|---------------------------|
| ANT        | -          | -           | Solder antenna here       |
| DATA       | Pin 20     | GP15        | Transmit signal           |
| VCC        | Pin 40     | VBUS (5V)   | Power (5V for high power) |
| GND        | Pin 38     | GND         | Ground                    |

#### SRX882 Receiver (Rectangular Board)

| SRX882 Pin | Pico Pin # | Signal Name | Purpose |
|------------|------------|-------------|---------|
|------------|------------|-------------|---------|

|              |             |          |                              |
|--------------|-------------|----------|------------------------------|
| Pin 1 (ANT)  | -           | -        | Solder antenna here          |
| Pin 2 (GND)  | Pin 38 or 8 | GND      | Ground                       |
| Pin 3 (VCC)  | Pin 36      | 3V3(OUT) | Power (3.3V for GPIO Safety) |
| Pin 4 (CS)   | Pin 36      | 3V3(OUT) | Active Enable (Tie to VCC)   |
| Pin 5 (DATA) | Pin 19      | GP14     | Receive signal               |
| Pin 6 (GND)  | -           | -        | Optional Ground              |

## 4. Solderless Connection Methods

If you do not have a soldering iron, you can establish physical connections using these tricks:

- Dupont Jumper Wires:

If your modules have header pins pre-soldered, push a female-to-male Dupont jumper wire directly onto the pin, and insert the male end into the breadboard.

- Wire Wrapping (Bare Holes):

Strip 1 cm of insulation off a solid core wire. Feed it through the board hole (e.g. ANT or DATA) and wrap it tightly 3-4 times around the metal-plated ring to make firm contact. Secure it with electrical tape or a drop of hot glue.

- Antenna Alternatives:

Use a standard 20 cm jumper wire as the antenna. If using a female-to-male wire, clip the female end onto the ANT pin header (if present). If the module has only a bare solder pad, wrap a stripped wire firmly and tape it. A length of 17.3 cm is ideal, but a standard 20 cm jumper works very well.

## 5. Range & Noise Optimization

To get the maximum range out of your transmitters and receivers, follow these physical rules:

1. Antenna Geometry:

Keep the 17.3 cm antenna wire as straight and vertical as possible. Do not coil it or place it parallel to other metal objects, ground planes, or large power cables.

2. Power Rail Filtering (Decoupling):

Microcontrollers like the Pi Pico generate electrical noise on their power rails. Solder or place a 100uF electrolytic capacitor and a 100nF ceramic capacitor in parallel directly across the VCC and GND columns where the receiver is connected. This smooths out voltage spikes.

3. Physical Separation:

If you ever migrate this to a full-sized Raspberry Pi 4 or 5, remember that their processors and USB 3.0 ports emit electromagnetic noise. Keep the receiver module at least 15-20 cm (6-8 inches) away from the Pi board using extension jumper wires.